Final Recommendation Options Logic

AID-LOG-010 | Revision 1

1. Purpose

This standard defines how AIDOOi converts the eligible, ranked products produced by AID-LOG-009 into complete customer-facing recommendation compositions. It governs composition quality, four distinct recommendation objectives, budget behaviour, and option diversity.

> **A final recommendation is a coordinated composition, not a collection of individually high-scoring products.**

2. Recommendation Architecture

AID-LOG-009 ranks individual products. AID-LOG-010 evaluates how selected products work together as a complete room or project recommendation. Products must remain compliant with all applicable Room, Project, CAT, DATA and LOG standards.

3. Four Final Recommendation Options

Option	Primary Objective
Best Overall Match	Default recommendation. Highest-quality complete composition for the customer as a whole.
Best Value	Maximize overall recommendation quality relative to budget spent; not the cheapest composition.
Best Style Match	Maximize visual and stylistic coherence while preserving fit, function and budget feasibility.
Premium Choice	Provide a meaningful quality/design upgrade with a controlled budget stretch of up to 20% above the selected budget ceiling.

4. Best Overall Match

Best Overall Match is the Primary and Default Recommendation. Decision priority: Fit & Function → Overall Match Quality → Composition Harmony → Budget → Availability. It is not simply the combination of the highest-ranked individual products; a slightly lower-scoring product may be selected when it creates a materially stronger complete composition.

5. Best Value

Best Value optimizes the complete composition for quality relative to money spent. Hard requirements and the LOG-009 minimum product quality remain unchanged. Savings on one product may be reallocated to another product when this improves the complete recommendation. AIDOOi must not add or upgrade products merely to consume unused budget.

> **Best Value maximizes overall recommendation quality per budget spent; it does not minimize total price.**

6. Best Style Match

Best Style Match prioritizes Style Match, Colour and Palette Harmony, Material Compatibility and overall design coherence after Fit and Function are satisfied. It must remain within the customer's selected budget ceiling and must not sacrifice mandatory requirements for visual preference.

7. Premium Choice

Premium Choice may extend up to 20% above the upper limit of the customer's selected budget. The additional spend must produce a meaningful improvement in product quality, material, design match or overall composition. Higher price alone is not an upgrade. If no meaningful improvement exists, AIDOOi must not manufacture an artificial premium option.

8. Meaningful Difference Rule

Final options should be meaningfully distinct where suitable inventory allows, but diversity must never override recommendation quality. AIDOOi should prioritize Anchor Product diversity across the four options. If inventory

quality or fit constraints make this impractical, an Anchor Product may repeat while supporting products and the optimization objective create a genuinely different composition.

9. Composition Score

Composition Factor	Weight
Product Match Quality	40%
Composition Harmony	30%
Functional Completeness	20%
Budget Performance	10%
Total	100%

LOG-009 scores products. LOG-010 scores complete compositions. Spatial eligibility and hard fit are protected upstream by the applicable Room Logic and AID-LOG-009. For MVP, Composition Score uses measurable structured data rather than claiming unrestricted spatial or aesthetic understanding.

10. Product Match Quality - 40%

Product Role	Weight within Product Match Quality	Examples
Anchor Product	50%	Sofa; Bed; Dining Table
Core Supporting Products	35%	Accent Chair; Coffee Table; Nightstand; Dining Chairs
Secondary / Decorative Products	15%	Lighting; Rug; Cushion; Decor

Each role uses the average AID-LOG-009 Match Score of products assigned to that role. If a valid composition does not require one role group, its weight is normalized across the groups that are present so compact or intentionally lean spaces are not penalized. Product roles are derived from the applicable composition logic in AID-LOG-002 through AID-LOG-008.

11. Composition Harmony - 30%

Harmony Factor	Weight
Style Harmony	50%
Colour Harmony	30%
Material Compatibility	20%

Composition Harmony evaluates products against each other, not only against the customer. For MVP, Style, Colour and Material relationships must use controlled compatibility rules and structured tags. Harmony means compatibility, not sameness. Products do not need identical style, colour or material to form a coherent composition.

12. Functional Completeness - 20%

Condition	Score
All Essential Functions plus appropriate Supporting functions covered	100
All Essential Functions covered; some useful Supporting elements absent	85
All Essential Functions covered; composition intentionally lean	70
An important / mandatory function is incomplete	Reject / Rebuild

Functional Completeness rewards appropriate functional coverage, not furniture quantity. Essential and Supporting functions are inherited from AID-LOG-002 through AID-LOG-008. A compact room or studio can achieve full functional completeness with fewer or multifunctional products.

13. Budget Performance - 10%

Condition	Score
Excellent Budget Use	100
Strong Budget Use	85
Acceptable Budget Use	70

Condition	Score
Weak / unbalanced allocation while still within budget	50
Over applicable budget ceiling	Reject / Rebuild

Budget Performance evaluates how intelligently budget is allocated across the complete composition. Spending less is not automatically better, and the system must not spend remaining budget unnecessarily. For Premium Choice only, the applicable ceiling includes the approved maximum 20% budget stretch.

14. Shared Quality Foundation and Optimization

Option	Optimization after quality foundation
Best Overall Match	Highest overall Composition Score.
Best Value	Strong Composition Score with superior quality-to-cost efficiency.
Best Style Match	Strong Composition Score with priority to Style, Colour and Material Harmony.
Premium Choice	Strong Composition Score plus a meaningful quality/design upgrade within the +20% ceiling.

All four options use the same underlying Composition Score and quality rules. The system should not create four unrelated scoring formulas. Each option applies a different optimization objective only after minimum quality is satisfied.

15. Minimum Composition Quality

Composition Score	Classification
90-100	Excellent Composition
80-89	Strong Composition
70-79	Good Composition
Below 70	Do Not Recommend / Rebuild

Minimum Composition Score = 70/100. In addition, every recommended composition must satisfy three hard rules: (1) no included product may have a LOG-009 Match Score below 60; (2) no Essential Function may be incomplete; and (3) no Fit or other Hard Requirement may be violated. These failures cannot be offset by a high average score.

16. MVP Implementation Boundary

For MVP, Composition Harmony is calculated from structured Style, Colour and Material compatibility; Functional Completeness from established room/project composition rules; Product Match Quality from AID-LOG-009; and Budget Performance from current prices and applicable budget logic. Advanced free-form spatial planning or visual aesthetic judgment is not required to implement this standard.

17. Output Handoff

The output of AID-LOG-010 is up to four complete, valid and meaningfully differentiated recommendation compositions: Best Overall Match, Best Value, Best Style Match and Premium Choice. Their visual presentation and product-faithful room/project rendering are governed separately by AID-VIS-001.

18. Final Principle

All four recommendation options share the same quality foundation. Their differences come from optimization objectives, not from lowering fit, functional or recommendation standards.